

RISC-V—A brief introduction

RISC-V (pronounced 'risk-five') is an open standard instruction set architecture (ISA) based on established reduced instruction set computer (RISC) principles. Unlike most other ISA designs, the RISC-V ISA is provided under open source licenses that do not require fees to use. Several companies are offering or have announced RISC-V hardware. Open source operating systems with RISC-V support are available and the instruction set is supported in several popular software tool chains.

The project began in 2010 at the University of California, Berkeley, but now many volunteers who are not affiliated with the university have started contributing. Unlike other academic designs, which are typically optimised only for simplicity of exposition, the designers ensured that the RISC-V instruction set be usable for practical computers.

As a RISC architecture, the RISC-V ISA is a load-store architecture. Its floating-point instructions use IEEE 754 floating-point. The instruction set is designed for a wide range of uses. The base instruction set has a fixed length of 32-bit naturally aligned instructions, and the ISA supports variable length extensions where each instruction could be any number of 16-bit parcels in length. Its subsets support small embedded systems, personal computers, supercomputers with vector processors, and data centre computers.

The instruction set specification defines 32-bit and 64-bit address space variants. The specification includes a description of a 128-bit flat address space variant, as an extrapolation of 32- and 64-bit variants.

As of June 2019, version 2.2 of the user-space ISA and version 1.11 of the privileged ISA are frozen, permitting software and hardware development to proceed. The user-space ISA, now renamed the Unprivileged ISA, was updated, ratified.

fabless vendors, who designed the best-performing processors and got them manufactured through outsourced semiconductor foundries, added more traction to the ARM based systems.

One interesting event that happened on sideline was the acquisition of mainline programmable logic device vendors, mainly FPGA (field programmable gate array) vendors like Xilinx and Altera, by mainline processor vendors. Intel bought Altera and AMD bought Xilinx. These acquisitions changed the game further.

These vendors acquired the programmable logic vendors to bolster their datacentre products as the datacentre interconnects were reaching 100Gps speed and there was need for hardware based solutions, which typically only FPGAs can offer! When things were coasting along, NVIDIA announced the takeover of ARM. (There are some issues which the Competition Commission is looking at, but as of now ARM is owned by NVIDIA.)

It is not that low-end 8- and 16-bit microcontrollers business

was hale and healthy. There was a huge churn in that market too, but the controllers made by different companies have not been dropped (so far!). Microprocessors like 68000 family and 8086 family have become obsolete. In the 8-bit controller side the situation is pretty bad! Processors like 65C02 from Rockwell, 6800 and 6809 from Motorola, 8085 processor and 8051 microcontroller from Intel have all become obsolete.

Interestingly, 65C02 is still being offered by some vendors as a custom ASIC (application specific integrated chip). 8051 does have incarnations from vendors like Atmel, Silicon Labs, and Nuvoton, who have processors supporting 8051 instructions but through their own custom IC architecture. Today 8- and 16-bit controllers are pretty much dominated by TI, Microchip, and NXP, who have significant market share. But this is not an exhaustive list. The idea is to set the context on what it takes to develop a microcontroller in the current situation.

With ARM being acquired by NVIDIA, one big challenge facing the ARM processor vendors is

that they do not have access to a vendor-neutral processor (where the core vendor is not associated to any major vendor but is independent), making the situation dicey. How NVIDIA will handle the ARM core licensing going forward is not known. So far the vendors are safe producing their existing products, but future will depend on what NVIDIA decides.

This has forced companies and countries to start worrying about the impact of these acquisitions and consolidations. With electronics taking over most of the day-to-day life, the equipment are getting critically impacted. This is especially true in the case of defense equipment. Independence of the technology, and even freedom to develop, will be under pressure due to access to advanced processor architecture and even the IP core, going forward!

Strategic issues

One of the biggest issues that everyone is facing is, how to develop products and solutions without any restrictions. Under this circumstance the only way to find a solution is to look at open source solutions. We in India have a successful track record of using Linux. With Linux taking over the internet servers and other critical applications (have to admit Linux couldn't replicate the success in the desktop market, barring the developer communities) we have demonstrated that open source technologies can be successfully adopted.

The logical thinking is to replicate the same model for processors. This has become feasible with the availability of RISC-V, an open source architecture which basically defines the instruction set architecture, leaving the implementation to the designers. This has led to multiple versions of RISC-V implementations available for designers to develop.

One advantage of the RISC V